# Assignment 8: Basic SQL Queries and Aggregate Functions

Database: MySQL | Project: Airline Reservation System

## 1. Objective

Demonstrate WHERE, ORDER BY, GROUP BY, HAVING and the COUNT, SUM, AVG, MIN and MAX aggregate functions.

## 2. WHERE — filter flights

```
SELECT flight_id, flight_number, departure_airport
FROM Flights
WHERE departure_airport = 'Bengaluru';
```

| flight_id | flight_number | departure_airport |
|---|---|---|
| 301 | 6E101 | Bengaluru |
| 304 | 6E404 | Bengaluru |

Returns flights departing from Bengaluru.

## 3. ORDER BY — sort payments descending

```
SELECT payment_id, booking_id, amount
FROM Payments
ORDER BY amount DESC;
```

| payment_id | booking_id | amount |
|---|---|---|
| 505 | 405 | 18000.00 |
| 503 | 403 | 12500.00 |
| 501 | 401 | 5500.00 |
| 507 | 407 | 5100.00 |

Lists payment amounts from highest to lowest.

## 4. GROUP BY — count bookings per flight

```
SELECT f.flight_number, COUNT(b.booking_id) AS total_bookings
FROM Flights f LEFT JOIN Bookings b
ON f.flight_id=b.flight_id
GROUP BY f.flight_id, f.flight_number
ORDER BY f.flight_id;
```

| flight_number | total_bookings |
|---|---|
| 6E101 | 2 |
| AI202 | 1 |
| IX303 | 2 |
| 6E404 | 3 |

## 5. HAVING — flights with at least two bookings

```
SELECT f.flight_number, COUNT(b.booking_id) AS total_bookings
FROM Flights f JOIN Bookings b ON f.flight_id=b.flight_id
GROUP BY f.flight_id, f.flight_number
HAVING COUNT(b.booking_id) >= 2
ORDER BY total_bookings DESC;
```

| flight_number | total_bookings |
|---|---|
| 6E404 | 3 |
| 6E101 | 2 |
| IX303 | 2 |

## 6. Aggregate functions

| Function | SQL expression | Expected output |
|---|---|---|
| COUNT() | SELECT COUNT(*) FROM Passengers; | 6 |
| SUM() | SELECT SUM(amount) FROM Payments; | 66000.00 |
| AVG() | SELECT ROUND(AVG(amount),2) FROM Payments; | 8250.00 |
| MIN() | SELECT MIN(amount) FROM Payments; | 4800.00 |
| MAX() | SELECT MAX(amount) FROM Payments; | 18000.00 |

SUM(amount) represents the total value of the payment records, including the pending payment.

## 7. Booking summary

```
SELECT COUNT(*) AS total_bookings,
       COUNT(DISTINCT passenger_id) AS unique_passengers,
       COUNT(DISTINCT flight_id) AS booked_flights
FROM Bookings;
```

| total_bookings | unique_passengers | booked_flights |
|---|---|---|
| 8 | 6 | 4 |

## 8. Conclusion

Filtering, sorting, grouping and aggregate functions support analysis of flights, bookings, passengers and payments.